D2 - Data and Assumption Register

MIN E 413 Term Project - Pit I HG stockpile reconstruction

This register is the controlled record of (1) the data received, (2) the quality findings, (3) every preprocessing and cleaning decision, and (4) every engineering assumption, with its need, justification, consequence and treatment. Parameter values live in `config/project.yaml`, and each value there is tagged with the ID used here (`A-xx`). Machine-readable versions of the QA/QC results are in `outputs/qaqc/`. **No raw record was deleted or edited.** Raw files are read in place from `Data_Topo 2026.zip` and identified by SHA-256 (`outputs/audit/data_registry.csv`).

Information type key, used throughout the project (course outline, Task 7): **M** = measured, **I** = inferred from data, **A** = assumed, **G** = model-generated.

1. Dataset inventory

ID	File(s)	Content	Coverage	Type	Notes
DS-1	`Dump Data.csv`	275 FMS truck-dump records: Cycle ID, payload (t), Au (g/t), Ag (g/t), Cu (%), density (t/m³), dump X/Y/Z, timestamp (minute resolution)	2026-03-15 17:43 to 2026-03-31 20:52; 44,650 t	M	No truck ID or source (pit/bench) field supplied. Grades are per **source signature** (21 distinct Cu/Au/Ag/density combinations), not per-load assays
DS-2	`Topo_260315.dxf` … `Topo_260327.dxf` (8 files)	Vulcan DXF, 3DFACE triangles, one layer per file	2026-03-15 to 03-27 (no 03-17, 03-23 to 03-26)	M	Each file is a **closed change solid**: a top surface (this survey) and a bottom surface (previous surface) over the area that changed (QA-S01)
DS-3	`Topo_250328.dxf`, `Topo_250329.dxf`	As DS-2	Re-dated to 2026-03-28 and 2026-03-29 (QA-S02)	M	File-name year inconsistent with the record
DS-4	(not supplied)	Full stockpile surface, pre-stockpile ground, density tests, dozer logs, truck IDs, moisture	None	n/a	Absence drives assumptions A-12, A-13, A-14, A-22. Requested as recommendations (D6)

Units and coordinate reference: metres. Easting is about 511,300 and Northing about 7,122,300, consistent with a UTM-type grid in the southern hemisphere. Elevation is about 4,480 m. DXF `$INSUNITS` is 6 (metres). FMS and surveys overlay without offset (dumps sit on and behind the surveyed crest), so **no transformation** was applied (A-01).

2. QA/QC findings

2.1 FMS records (DS-1)

ID	Finding	Evidence	Action / treatment	Influence on model	Traceability
QA-F01	No duplicate IDs, missing fields or unparseable timestamps	Flags F01-F03 = 0	None	-	`fms_flag_summary.csv`
QA-F02	**Payload is not weighed.** All 275 payloads are exact multiples of 38 t (38-304 t; 152 t = 155 loads), so payload is a pass-count estimate (bucket factor × passes)	`payload_distribution.csv`	Retained. Per-load payload error CV 8% plus a ±3% systematic bias per Monte Carlo realisation (A-10). 2 part loads (38, 76 t) and 3 heavy loads (266, 304 t) flagged F05/F06	Tonnage, and therefore metal, uncertainty	Flags kept in `fms_records_with_flags.csv`
QA-F03	**Grades are source-block values**, not load assays: 21 signatures, each repeated across many loads. Cu 5.6-13.7%, Au 5.6-10.1 g/t, Ag 147-256 g/t, density 1.92-2.22 t/m³	`grade_signatures.csv`	Treated as the best available grade per load. Grade error modelled per signature (correlated across loads of the same source), CV 10% (A-11)	Within-load homogeneity assumed (A-11)	Signature ID `grade_group` added
QA-F04	Grade magnitudes are very high for a Cu-Au mine (mean 8.4% Cu, 8.0 g/t Au). Possible unit scaling or anonymisation	Summary statistics	Units used **as supplied** (% Cu, g/t Au/Ag), labelled in every output. Flagged for confirmation with the mine geology team	Absolute metal values depend on unit confirmation; relative patterns do not	Recommendation R-07

ID	Finding	Evidence	Action / treatment	Influence on model	Traceability
QA-F05	33 Cycle IDs out of chronological order	F04	Information only. The timestamp defines order	None	Flag
QA-F06	**Dump Z is unreliable**: Z minus surveyed surface at dump time has median −0.7 m and P10/P90 −2.8/+2.0 m (n = 200). 14 dumps lie below the fitted floor and 29 have |ΔZ| > 3 m	`fig_qaqc_dumpz_timeofday.png`	Z not used by the algorithm. Only X/Y is used (A-29)	Removes a noisy input	F10/F11 flags
QA-F07	**FMS completeness is uneven in time.** No records for 2026-03-17, 03-23, 03-24, 03-26, 03-27 or 03-30. Single records on 03-25 and 03-31. Records stop at 12:27 on 03-16 and restart at 20:47 on 03-18 (56 h silence) while the surveys show 4,843 m³ placed	`daily_coverage.csv`, `fig_qaqc_daily_tonnage_volume.png`	Not imputed as loads. The observed volume not explained by FMS becomes **unattributed** material with an imputed grade and a separate class (A-21)	Large unattributed share in periods 03-18 and 03-27	Period table; class fields in the block model
QA-F08	FMS starts at 17:43 on 03-15, part-way through the first survey period	`daily_coverage.csv`	Period 0 treated as partially covered: 65% of its fill is unattributed	As above	

ID	Finding	Evidence	Action / treatment	Influence on model	Traceability
QA-F09	1 load (228 t, 03-31 20:52) is after the last survey	F12	Placed **provisionally** with the forward physics model and flagged; re-reconciled automatically when the next survey arrives	0.1% of tonnes	Class “Provisional”
QA-F10	All dumps lie within 60 m of surveyed change (F09 = 0); 18 dumps fall in survey-gap periods (F13)	Flags	Retained	-	Flags
QA-F11	**Default / stuck FMS position.** 14 records (2,356 t, 5.3% of FMS tonnes, 03-16 to 03-28) share the identical position E 511,334.7, N 7,122,320.4, Z 4,473.0. That position is below the floor and, until 03-27, in front of the face toe. These are also the 14 records of flag F10	`fms_records_with_flags.csv` (F14)	Retained. X/Y imputed as the tonnage-weighted centroid of the other dumps of the same period (A-32); original X/Y kept (`x_raw`, `y_raw`); GPS error ×5 in Monte Carlo	Avoids placing 2.4 kt from a fictitious point at the toe	F14 flag, `loc_imputed` column in `movement/loads_used.csv`

2.2 Surveys (DS-2, DS-3)

ID	Finding	Evidence	Action / treatment	Influence
QA-S01	**Each DXF is a closed change solid, not a surface.** 100% of cells are crossed by exactly two faces. The divergence-theorem volume equals the gridded volume (e.g. 2,868 vs 2,867 m³). The bottom of solid *s* equals the top of solid *s−1* wherever they overlap (P5-P95 = 0.000 m)	`survey_qaqc.csv`	Top = surface at survey *s*; bottom = surface at the previous survey; only changed areas are present	Defines fill per period directly (M). The pre-record surface is observed only where change occurred
QA-S02	File names `Topo_250328`/`Topo_250329` carry year 2025. All ten layer names use year '25' (e.g. layer `250315_STOCK_HGHS` in `Topo_260315.dxf`). FMS shows dumping on 2026-03-28/29. The faces advance beyond the 2026-03-27 solid, and their bottoms match its top	`survey_qaqc.csv`, `fig_growth_face_advance.png`	Re-dated to 2026-03-28 and 2026-03-29 via `config/survey_overrides.csv` (manual intervention, reason recorded, applied in code not by renaming files)	Chronology
QA-S03	**Survey timing not supplied.** Correlation of period FMS volume with surveyed volume is r = 0.96 if surveys represent end of day (23:59), and r ≤ 0.21 for any time before 10:00	`temporal_alignment.csv`, `fig_temporal_alignment.png`	Survey time = 23:59 of the file date (A-05); 21:00-23:59 sampled in Monte Carlo	Assignment of loads to periods
QA-S04	Missing surveys: 03-17 and 03-23 to 03-26. Bottoms of the 03-18 and 03-27 solids differ from the previous tops on 23% and 14% of overlapping cells	`survey_qaqc.csv`	Periods flagged as gap periods; unobserved fill on first-surveyed cells inferred (A-13b)	Larger uncertainty in those periods
QA-S05	**Elevation decreases are not stored as solids.** 22% of overlap cells in the 03-18 solid and 6% in the 03-28 solid have bottoms up to 1-2 m *below* the earlier surveyed surface (dozer cuts or slumping)	`period_reconciliation.csv` (`frac_lowered`, `cut_volume`)	Treated as cuts: the top of the column is removed and re-handled as a transport source (A-20). Total 567 m³	Mixing of earlier material into later fill
QA-S06	4 surveys (03-16, 03-21, 03-22, 03-29) do not reach the floor (minimum Z 4,476.4-4,478.8 m vs floor about 4,475 m): the change polygon was clipped above the toe	`reaches_floor`	Toe fill captured later is attributed by A-13b	Part of the 7,232 m³ of "unobserved fill"
QA-S07	`Topo_250328.dxf` has inverted face winding (signed volume −4,924 m³). There are a few degenerate (≤ 25) and duplicate (≤ 6) triangles per file	`survey_qaqc.csv`	Rasterisation is winding-independent (top = max, bottom = min); degenerate faces are skipped	None

ID	Finding	Evidence	Action / treatment	Influence
QA-S08	Triangle edge median 0.93-1.15 m, so the native resolution is about 1 m	`edge_median_m`	Simulation grid 1 m (A-24)	Resolution
QA-S09	Face slopes (cells > 25°, n = 13,084): median 36.7°, mode 37.0°, P75 38.5°, P90 40.1°	`fig_qaqc_repose.png`	Angle of repose 37° (A-03)	Forward model, inference
QA-S10	Toe elevations define a near-horizontal floor: plane 4,475.08 m, dip 0.29°, residual sd 0.26 m (n = 197)	`fig_qaqc_floor_plane.png`	Floor plane (A-12)	Legacy thickness
QA-S11	Survey precision cannot be estimated from repeat surveys (bottoms are copies of previous tops). 8-16% of solid area is thinner than 0.1 m	`sliver_frac_lt_0.1m`	σz = 0.05 m assumed; detection limit τ = 0.14 m; changes below τ are not moved (96 m³ in total)	Negligible volume

2.3 FMS vs survey reconciliation (Task 2 summary)

The surveyed change solids total **35,096 m³**. With the inferred unobserved toe fill (A-13b), the observed fill used by the model is **42,353 m³**. FMS tonnage converts to **21,640 m³** at the FMS densities. In periods with complete FMS and survey coverage, the apparent density (FMS t ÷ solid m³) is 1.2-2.8 t/m³ day by day, with a pooled value of 1.67 t/m³ (39,596 t / 23,708 m³), against a mean FMS density of 2.06 t/m³. The day-to-day scatter is dominated by dozer timing (material dumped late in one survey day and pushed the next). For example, 03-20 shows 345 m³ more FMS volume than fill; that volume is carried to 03-21. The pooled deficit (FMS volume = 81% of surveyed solid volume on the “complete” days; 56-131% day by day) has two non-exclusive explanations that the supplied data cannot separate: (a) swell, so the in-pile density is lower than the FMS density (κ < 1); (b) unrecorded loads. Both are carried as uncertainty (A-06, A-21).

3. Preprocessing decisions

ID	Decision	Why necessary	Action	Possible influence	How original is retained
PP-01	Survey re-dating	Chronology	Override table with reason and approver	Wrong order would corrupt sequencing	Original names unchanged; `date_from_name` and `assigned_date` both stored
PP-02	Survey time 23:59	Assign loads to periods	Config value, data-driven choice	±1 period for loads near midnight	Alignment table
PP-03	Rasterise solids to 1 m grid (top/bottom at cell centres)	Common computational grid	Cached by file hash	≤ 1 m edge effects	Raw DXF untouched
PP-04	Dump Z ignored	Unreliable	Only X/Y used	None	Z kept in loads table
PP-05	Unattributed fill volume	FMS gaps	Separate material class, imputed grade	Grade of 21% of tonnes is imputed	Class tonnage per block

ID	Decision	Why necessary	Action	Possible influence	How original is retained
PP-06	Cuts detected from bottoms below earlier tops	Solids omit decreases	Removal + re-handling	Mixing	`cut_volume` per period
PP-07	Load after last survey	Not yet surveyed	Forward-model nowcast, flagged	228 t	Class “Provisional”
PP-08	No record excluded	Brief: do not simply delete	Flags only	-	All 275 records used
PP-09	Location of 14 default-position records imputed	Position is physically impossible (QA-F11)	Period centroid of other dumps	Their material is allocated from the platform instead of the toe	`x_raw`/`y_raw` retained; flag F14

4. Assumption register

Columns follow the course outline, Task 4: Assumption | Need | Justification | Consequence if wrong | Treatment. **S** = tested in sensitivity (one-at-a-time) and/or **MC** = sampled in the Monte Carlo uncertainty analysis.

ID	Assumption (value / rule)	Need	Justification	Consequence if wrong	Treatment
A-01	Coordinates of FMS and surveys share one projected mine grid (m); no transformation	Overlay data	Dumps lie on and behind surveyed crests; consistent elevations	Systematic offset would bias push distances	Visual + statistical check (QA-F06, F09)
A-02	`Topo_2503xx` files are 2026 surveys	Chronology	QA-S02 evidence (FMS dates, face progression, bottom = previous top)	Sequence error	Manual override logged
A-03	Angle of repose 37°	Forward model; inference of unseen faces	Measured mode 37.0°, median 36.7° of surveyed faces; typical 35-40° for blasted rock	Shape of inferred faces and nowcast	S (forward calibration 35/37/39°)
A-04	DXF solids record accretion only; bottom = previous surface	Derive fill per period	QA-S01, QA-S05	Misread change	Verified on all 9 overlaps
A-05	Surveys represent the stockpile at 23:59 of the file date	Assign loads to periods	Alignment r = 0.96 vs ≤ 0.21 for morning (QA-S03)	Loads placed one period early or late	S (21:00 vs 23:59), MC (21:00-23:59)
A-06	In-pile bulk density = FMS density × κ, κ = 1.0	Convert tonnes to volume	FMS density is the mine’s own value (M); calibrating κ from volumes is confounded with missing loads (Section 2.3)	Tonnage of unattributed/legacy volume and FMS footprint size change proportionally	S (0.90/1.10), MC U(0.80, 1.05)

ID	Assumption (value / rule)	Need	Justification	Consequence if wrong	Treatment
A-07	Truck GPS horizontal error σ = 3 m	Uncertainty	Typical autonomous/HP GPS on haul trucks 1-5 m; dump-event latency	Destination of individual loads	MC
A-08	Survey vertical precision σz = 0.05 m	Detection limit, uncertainty	Typical drone photogrammetry/LiDAR on rock piles 0.03-0.10 m	Noise counted as fill	S (0.025/0.15), MC (per-survey bias 0.05 m)
A-09	Detection limit τ = 2·√2·σz = 0.14 m	Ignore noise	Standard LoD for DEM differencing (95%)	96 m³ not moved	Robustness (k = 1 to 4)
A-10	Payload error CV 8% per load, ±3% systematic	Tonnage uncertainty	Pass-count payload (QA-F02); industry payload-monitor accuracy ±5-10%	Tonnes and metal	MC
A-11	Grade of a load = its source signature; homogeneous within a load; signature error CV 10%	Assign grade	Only data available (QA-F03)	Block grade variance under-represented	MC (correlated by signature)
A-12	Floor = plane fitted to toe cells (4,475.08 m, 0.29° dip)	Bottom of model	Toe elevations consistent across surveys (sd 0.26 m)	Legacy volume ±0.26 m × area (about ±2.6 kt per 5,000 m²)	Reported; floor residual map
A-13	Never-surveyed cells: behind the crest = flat platform at the nearest crest elevation; in front of the face = face continued at 37° to the floor	Geometry where no survey exists	Truck-operated platform is flat; dumps on platform; faces at repose	Legacy volume/extent	Geometry class “inferred” in block model; legacy reported separately
A-13b	First-surveyed cells in front of an earlier surveyed face were at the repose extrapolation of that face; a solid bottom above it is fill placed during the record but not captured by a solid (toe-clipped solids, missing surveys)	Avoid counting recent fill as legacy	QA-S04, QA-S06: toe-clipped solids; solid bottoms 2-4 m above the floor directly in front of earlier faces	7,232 m³ moves between “legacy” and “unattributed”	S (switch on/off)
A-14	Model domain = convex hull of surveyed change and dump locations	Lateral limit of the pile	No full-pile survey supplied	The back of the pile outside the hull is not modelled	Stated limitation; R-01

ID	Assumption (value / rule)	Need	Justification	Consequence if wrong	Treatment
A-15	Dozer push direction/destination follows least total push work (optimal transport over a slope-aware geodesic cost)	Unknown dozer trajectories	Operators minimise push distance; OT is the classical earth-moving formulation (Monge, 1781)	Wrong allocation of grade inside a period's fill	MC/S on ε, α, Lmax; twin experiment
A-16	Uphill push penalty α = 4 m per m of lift	Prevent physically implausible uphill pushes	Dozer production falls sharply when pushing uphill (grade correction factors, Caterpillar Performance Handbook)	Allocation across the crest	S (1/10), MC U(1, 10)
A-17	Maximum effective push distance 90 m (geodesic)	Limit movement	Dozing is generally considered economical to about 100 m (Caterpillar Performance Handbook); see the reconstructed push-distance distribution	Loads may be forced to platform (retained)	S (50/130), MC U(50, 130)
A-18	Mixing length ε = 6 m (entropy regularisation)	Degree of mixing	Combination of GPS error (3 m), blade spill, rolling on the face; twin experiment favours small-moderate ε	Over- or under-mixing of grades	S (2/15), MC U(2, 15), twin
A-19	Within one period, the fill of a column is vertically homogeneous	Vertical placement	Daily fill is built by many interleaved loads	Vertical grade resolution limited to the period layer (0.5-3 m)	Inherent; documented
A-20	Elevation decrease below an earlier surface = material removed by dozer/slumping and re-handled in the same period	Solids omit cuts (QA-S05)	Dozers trim crests while pushing	567 m^3 re-mixed	Reported per period
A-21	Unattributed fill has the tonnage-weighted mean grade/density of FMS loads in the same and adjacent periods	FMS gaps (QA-F07)	Material placed in the same days most likely came from the same faces	Grade of 41.8 kt is imputed	MC bootstrap of grades; separate class
A-22	Legacy (pre-record) material: density and grade = FMS mean (8.42% Cu, 8.00 g/t Au, 2.06 t/m^3)	Populate blocks below the pre-record surface	No information supplied	116 kt carry an **assumed** grade	Class "Legacy-assumed", confidence "Uninformed"; never reported as measured

ID	Assumption (value / rule)	Need	Justification	Consequence if wrong	Treatment
A-23	FMS volume not seen in that period's fill is retained on the platform and pushed in a later period (dozer lag)	Timing mismatch	Paddock dumping and pushing later is normal practice	Grade appears in the next period's fill	Carried volume reported
A-24	Simulation grid 1 m, vertical integration step exact (layers)	Numerics	Matches survey triangle size (QA-S08)	Edge effects	Robustness 0.5/2.5 m
A-25	Block size 5 × 5 × 2 m, partial blocks kept with fill fraction	Planning unit	Information resolution about 5 m (GPS + mixing). A 2 m vertical gives several layers per face height; small enough for loader reclaim slices	Smoothing of grade	S (2 × 2 × 1, 10 × 10 × 4)
A-26	Tonnes as reported by the FMS; no moisture correction (wet/dry basis unknown)	Units	No moisture data	Systematic tonnage bias of a few %	Stated; R-05
A-27	No particle-size segregation of grade down the face	Not observable	No size-grade data	Toe vs crest grade bias	Stated limitation; R-06 sampling
A-28	No time-dependent settlement after placement	Not observable	Short record (2 weeks)	Small volume bias	Absorbed in κ range
A-29	Dump Z is not used	QA-F06	Unreliable	None	-
A-30	Boundary conditions: material cannot leave the domain; outside the domain is not modelled	Numerics	Domain covers all observed change	None inside the domain	-
A-32	Records sharing an identical X/Y/Z with ≥ 2 others carry a default/stuck FMS position; their dump X/Y is the tonnage-weighted centroid of the period's other dumps	14 records have an impossible location (QA-F11)	Identical coordinates to 0.01 m over 12 days, below the floor, cannot be real dump events	Allocation of 2.4 kt within its period	GPS σ ×5 for these loads in MC
A-31	Missing data: FMS gaps → unattributed class; survey gaps → gap periods; no synthetic loads are invented	Missing-data treatment	Transparency over completeness	Unattributed share is large (21% of tonnes)	Reported; R-02

5. Data that would most reduce uncertainty (value of information)

1. A full-pile survey (current) and a pre-stockpile ground survey (removes A-13, A-14 and most of A-22).
2. FMS completeness audit for 03-16 to 03-18 and 03-22 to 03-28 (removes most unattributed volume).
3. Bulk density and moisture tests on the stockpile (A-06, A-26).
4. Truck ID, source bench and per-load grade (A-11).
5. Tracer trials (RFID or coloured tags in selected loads) to calibrate ε and the push model (A-15 to A-18).